

Vehicle Detection for Indian Urban Roads

YOLOv8 + IISc AIM UVH-26

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Faculty progress deliverable - 11 September 2026

26,646	316,220	14	47
Images listed	MV object boxes	Vehicle classes	Tests passed

Verified progress, with a clear experimental boundary.

Dataset annotation audit, EDA, data-pipeline implementation and a genuine one-epoch training smoke test are complete. Full image acquisition, full image integrity checks and the proper baseline remain in progress.

The problem

Indian traffic combines dense scenes, occlusion and fine-grained vehicle types. This project prepares a reproducible detector for categories such as two-wheelers, three-wheelers, LCVs and different passenger-car types.

Pipeline: pinned dataset -> audit -> COCO-to-YOLO conversion -> validation -> transfer learning -> evaluation.

Real-time analytics is the project objective. End-to-end real-time performance and final detection accuracy have not yet been established.

Inspect first. Preserve the splits.

Majority Voting split	Images listed	Object boxes
Training	21,349	252,723
Validation	5,297	63,497
Total	26,646	316,220

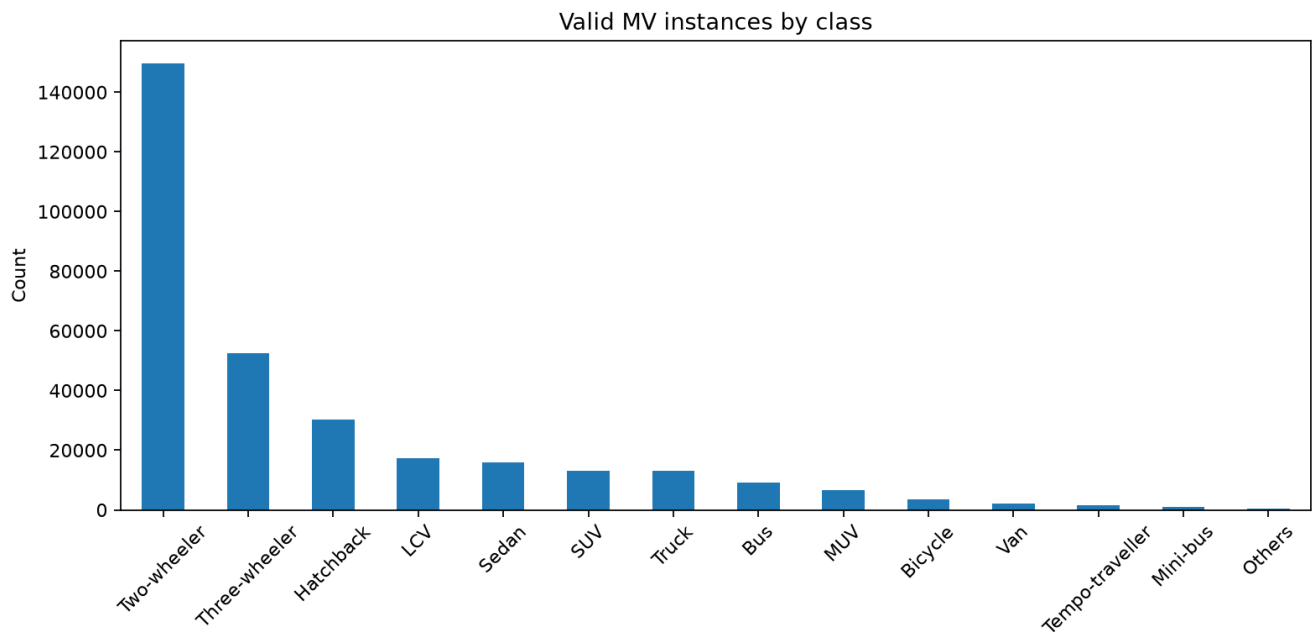
These are counts from the downloaded annotation JSON files. They do not imply that all image files are already present. MV is the selected ground truth; STAPLE is kept separate because its inspected files list 21,726 images and 283,402 objects.

All 14 original category names are preserved. Sorted original IDs 1-14 map to YOLO IDs 0-13.

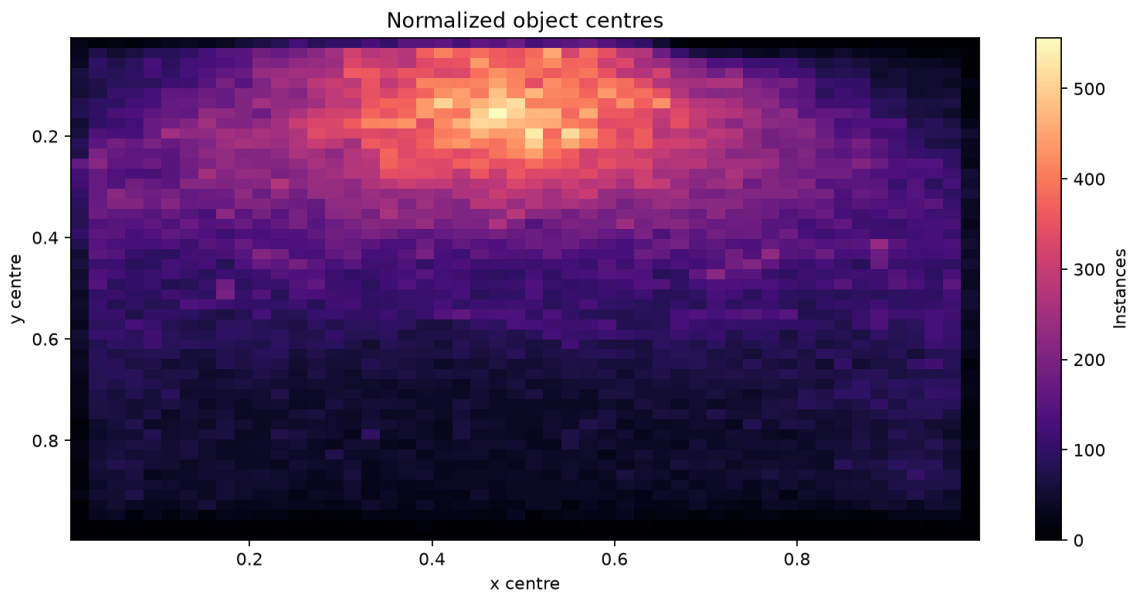
YOLO ID	Category	MV instances
0	Hatchback	30,290
1	Sedan	15,950
2	SUV	13,175
3	MUV	6,523
4	Bus	9,286
5	Truck	13,011
6	Three-wheeler	52,428
7	Two-wheeler	149,730
8	LCV	17,345
9	Mini-bus	873
10	Tempo-traveller	1,680
11	Bicycle	3,391
12	Van	2,186
13	Others	352

Annotation-only checks: zero invalid boxes, missing mandatory fields, duplicate IDs within each split or shared train-validation filenames. Full image decoding, dimension checks and content-hash leakage checks are still required.

Imbalance and crowded scenes



Two-wheelers: **149,730 instances**. Others: **352 instances**. Per-class reporting is essential; an aggregate score can hide poor rare-class performance.



Objects per image: mean **11.87**, median **10**, maximum **66**. At original resolution, 4,041 boxes are smaller than 32 x 32 pixels in area. The heatmap shows where annotated object centres occur; it is not a traffic-density prediction.

A real training smoke test

The 96-image pilot was independently decoded, dimension-checked and hashed across splits. Ultralytics scanned it with zero corrupt images. Training losses were finite, validation completed, and best/last checkpoints were saved.

Configuration	Executed value
Hardware	Apple M5 MacBook Pro, 24 GiB memory
Environment	Python 3.12.14; PyTorch 2.14.0; Ultralytics 8.4.146
Model / device	COCO-pretrained YOLOv8n / MPS
Pilot / epochs	64 training images + 32 validation images / 1 epoch
Image size / batch	640 / 8
Optimizer	AdamW; initial learning rate 0.000556
Total wall time	35.73 seconds

Standalone smoke validation - scores on a 0-to-1 scale

Precision	Recall	Macro F1	mAP50	mAP50-95
0.022367	0.002379	0.004019	0.000625	0.000433

This is not the proper baseline. The smoke checkpoint has poor detection quality. All 32 saved predictions had zero detections at confidence 0.01. The run proves that the training/evaluation pipeline executes; it does not establish final project accuracy.

47 tests passed. Synthetic tests cover geometry, class mapping, invalid inputs, empty labels, split manifests, reproducible selection and leakage blocking. MPS reported nondeterministic operations despite deterministic settings; exact numerical replay is not guaranteed.

What can be shown today

Show	Evidence
Data analysis	Audited annotation counts, class chart and spatial heatmap
Engineering	Versioned conversion code, immutable raw-data policy and tests
Executed training	Smoke epoch CSV, checkpoint checksum and per-class evaluation
Honest limitations	Missing/loose source labels, redacted imagery, unfinished full audit

A short offline demonstration

```
python3 scripts/show_progress.py
```

Then open the EDA figures and walk through the source code. With dependencies installed, run `python -m pytest -q`. The portable pack includes saved evidence; raw images and weights remain local.

Next: the proper baseline

The deterministic 8,000/2,000 train/validation selection covers all 14 classes. The maximum class-share deviation from the corresponding full split is **0.306 percentage points**. Its annotations contain 94,484 training objects and 24,342 validation objects. Acquisition and preparation are in progress.

Planned settings: YOLOv8n, 30 epochs, image size 640, batch 8, seed 42, MPS. Finish full-image audit, inspect the prepared subset, run the proper baseline, evaluate its best checkpoint, review errors and then complete the Git checkpoint.

Boundaries

No final accuracy or useful real-time deployment claim. No proper baseline commit or GitHub push has been completed. This deliverable is suitable for a progress review, not final project completion.

Sources and reproducibility

Dataset: [IISc AIM UVH-26](#), CC BY 4.0.

Revision: 59f82c57821e8a54dc40bc1f42e83909dbad0b70.

Citation: Sharma et al., IISc Technical Report, November 2025, [arXiv:2511.02563](#).

All project numbers above come from included audit, EDA, test and smoke-run artifacts.